

Model Integration & Performance Report

Project: Voice Agent Pro 2.0 - Enterprise Upgrade

Date: May 08-09, 2026

1. Executive Summary

This report documents the exhaustive integration testing and benchmarking phase conducted to achieve sub-second conversational latency. Multiple open-source and commercial stacks for Speech-to-Text (STT) and Text-to-Speech (TTS) were evaluated, with a specific focus on Indian code-switching (Hinglish/Marathi-English) and barge-in responsiveness.

2. Speech-to-Text (STT) Evaluation Log

Model Tested	Integration Type	Latency (Avg)	Dialect Accuracy	Verdict	Rationale for Failure
Whisper (Local/Faster)	Dockerized Container	1.8s - 2.5s	72%	FAIL	High CPU overhead caused cumulative latency. Failed to catch vernacular Marathi fillers.
Groq Whisper (Turbo)	REST API	0.8s - 1.2s	81%	FAIL	Severe hallucinations during silence. Standardized Hinglish into formal English, losing intent.
NVIDIA Canary	ONNX Runtime	1.1s	88%	FAIL	Required high VRAM (A10G+). Difficult to scale for 20+ concurrent

					calls without massive infra cost.
Google STT	gRPC Stream	1.5s	84%	FAIL	Native endpointing was sluggish. Failed to distinguish barge-in speech from background noise effectively.
Deepgram Nova-2	WebSocket	< 300ms	96%	PASS	Superior code-switching. Adaptive VAD handles Indian environments effectively. Sub-second turn-taking achieved.

STT Failure Analysis

- The "Hallucination" Trap: Most Whisper-based models generated irrelevant phrases such as "Thank you for watching" during silence or noisy environments.
- The "Standardization" Issue: Models such as Google STT and standard Whisper converted Indian vernacular into overly formal English, disconnecting the response tone from the user.

3. Text-to-Speech (TTS) Evaluation Log

Model Tested	Integration Type	TTFB (Avg)	Indian Accent Quality	Verdict	Rationale for Failure
Kokoro-ONNX	Local Python	400ms	35%	FAIL	Very robotic for Indian names and addresses. No support for Marathi

					phonemes.
Edge-TTS	WebSockets (Unofficial)	1.2s	65%	FAIL	No streaming generation. Required full sentence synthesis before playback.
ElevenLabs	WebSocket Stream	600ms - 900ms	92%	FAIL	Latency fluctuated globally. High cost-per-character made it unviable for mass-market SaaS.
Cartesia Sonic	WebSocket Stream	< 150ms	94%	PASS	Fastest TTFB observed. Native Indian English voices sounded natural and human-like.

TTS Failure Analysis

- The "Buffering" Wall: Edge-TTS and local models required the full text before starting audio playback, introducing dead-air gaps exceeding 1.5 seconds.
- Phonetic Struggle: Most models failed to pronounce local Indian area names accurately (e.g., Baner, Kothrud, Worli), reducing conversational immersion.